

Powering TV Experiences with Anytime Environmental Exploration

Celina Ma
University of Calgary
Calgary, Canada
celina.ma@ucalgary.ca

Haohong Wang
TCL Research America
San Jose, CA, USA
haohong.wang@tcl.com

Hao Sun
University of Missouri
Columbia, MO, USA
hshq7@mail.missouri.edu

Elliot van Huijgevoort
Cornell University
Ithaca, NY, USA
ejv37@cornell.edu

Mea Wang
University of Calgary
Calgary, Canada
meawang@ucalgary.ca

Zhihai He
University of Missouri
Columbia, MO, USA
hezhi@missouri.edu

ABSTRACT

Our work proposes a novel interactive model for TV such that users can freely decide when and how much to interact with the story. At any time, users can enter the immersive 3D environment of each scene in the story by controlling an avatar. In this scene, users embrace a rich set of interactive possibilities, including in-depth exploration, conversations with characters, and gamified quests to guide the story. Users also have the option to watch their avatar explore the 3D environment automatically for a lean-back experience. User evaluation of our prototype confirmed the acceptance of this new model and enthusiasm for the interactive freedom and depth it provides. To the best of our knowledge, this is the world's first lean-back-compatible TV model that offers a complete immersive experience driven by users, shedding light on a new direction for designing interactive videos appealing to a wide audience.

CCS CONCEPTS

• **Applied computing** → **Media arts**; • **Human-centered computing** → **Interaction design**.

KEYWORDS

interactive video, interactive narrative, environmental storytelling, gamification

ACM Reference Format:

Celina Ma, Haohong Wang, Hao Sun, Elliot van Huijgevoort, Mea Wang, and Zhihai He. 2021. Powering TV Experiences with Anytime Environmental Exploration. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451600>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451600>

1 INTRODUCTION

Traditional television (TV) and film are a "lean-back" pastime requiring no input from the user. In contrast, typical or virtual reality (VR) games let users influence their viewpoint and the narrative itself, but require users' full-time control. To balance lean-back viewing with active participation, VR films immerse the user in a 3D virtual environment - a digital representation of the story world [11]. By wearing a headset, the user is immersed in the film, and can look around the environment in 360 degrees. This environment supports gamification and offers a flexible interface for driving branching narratives. Users can play a more active role by performing diverse interactions, like finding key items or speaking to story characters. However, the duration of current VR stories is limited by the challenges of eye fatigue and motion sickness [6]. In addition, the wide adoption of VR is prevented by the cost of high-end VR headsets and the requirement of full-time wearable devices. This restricts VR entertainment to a niche market.

Our contribution focuses on wearable-free TV and film entertainment for the mass market. Interactivity has been explored in this field with interactive films like Netflix's *Black Mirror: Bandersnatch* [4]. Such stories allow the user to make choices at fixed points to determine story outcomes. These experiences were positively recognized by audiences, as they enjoyed impacting the story [21]. However, when compared to free control, fixed choices may limit immersion and perceived autonomy.

By adding the ability (seen in VR) to navigate a virtual environment to regular TV, a new interactive model is proposed. The model offers enhanced presence and freedom by placing an avatar representing the user in the film environment (earlier work can be found in [27, 28]). The avatar can examine details, speak with story-related agents, and play simple games. This avatar enables the environmental interactions seen in VR, without requiring a headset. We describe the original video as 2-dimensional (2D). Upon interacting with the 2D video at any time, the user can freely explore the current frame as a 3-dimensional (3D) virtual environment. This satisfies users who would like more interactive depth, while still being able to return to a normal TV show. The lean-back nature of TV is emphasized by having all interactions be voluntary, so users can decide how much to engage with the environment. Furthermore, we incorporate an automatic avatar exploration mode, enabled by reinforcement learning techniques. This benefits passive users, who

can watch the avatar interact with the environment without taking full control. In the remainder of this paper, we present related work in Sec. 2 and system design in Sec. 3. Our implementation and user evaluation through structured interviews are given in Sec. 4, and we conclude with our final thoughts.

2 RELATED WORK

Traditional books and films are consumed passively by users. However, interactive narratives offer users some degree of control over the story to enhance their engagement. Choose-your-own-adventure books were an early form of interactive storytelling, where the reader is prompted to make a choice and flip to a specific page to proceed. Different choices lead to alternate scenes, forming unique story paths. Full-motion video (FMV) games [18] and modern interactive films, including *Black Mirror: Bandersnatch* [4] and *Late Shift* [29], have followed this format of explicit choices. Another common interactive film model is letting the user switch between multiple viewpoints while watching [12, 30]. In these works, story events do not change, but the user’s interpretation can vary based on their viewing order.

These models have limited interactivity compared to game-like systems with exploratory user actions. Textual interactive fiction lets users freely type commands to examine and navigate the story environment [15]. More recent narratives extend on these textual origins with fully rendered virtual environments. Story-driven games like *The Wolf Among Us* [24] and *Until Dawn* [22] often couple branching choices with environments that users can explore freely. This creates a sense of discovery, as users play an active role in the story. Environmental details can drive narratives such as *Gone Home* [25] and the VR piece *Anonymous* [3], where the user examines a residence to learn about its occupant(s). Other VR stories focus strongly on role-playing due to the enhanced user immersion [8, 10].

The agency provided by game-like systems is rewarding, but usually demands the user’s complete engagement to progress. Therefore, it is not naturally inclusive for those who prefer passive viewing. VR films have experimented with gaze-based interactions that activate special events [16], or even drive branching storylines [20]. This maintains a lean-back experience, but some users may desire the option for more interactive freedom. Our film harnesses 3D environments for greater interactive depth, while letting users determine how much they wish to explore. This is similar to the work of Cavazza et al., who prototyped interactive narratives driven by autonomous characters [5, 13]. Like our system, the user can act as an unseen observer who manipulates the environment to influence the story at any time. We integrate this concept with a longer, human-written narrative to assure story quality and offer richer interactions.

3 SYSTEM DESIGN

In this section, we present the model’s design principles and system details. In contrast to a typical lean-back TV experience, the proposed system brings the following combination of benefits that does not exist in the current state-of-the-art: (1) Opportunities to switch between the original 2D video and interactive 3D scenes at any time; (2) Opportunities to explore and interact with the 3D

environment; (3) Opportunities to stay in lean-back mode to watch the avatar explore the environment on the user’s behalf.



Figure 1: The user avatar (left) in a 3D scene next to a story character (right).

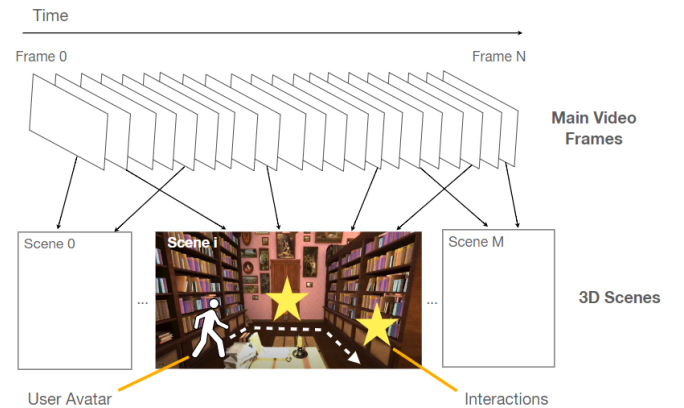


Figure 2: The anytime environment exploration model.

3.1 Anytime 2D/3D switching

Fig. 2 demonstrates the anytime environment exploration model. Video content is shown as a *timeline* of N frames at the top representing the TV watching experience. Associated 3D scenes are shown at the bottom, depicting the environment that each frame takes place in. The links between frames and scenes have a many-to-1 mapping relationship; some of the arrows transitioning toward the scenes cross each other, since a video may jump between environments several times. With this structure, the user can interact anytime to enter a rendered 3D environment corresponding to the current frame. Among the total of M 3D scenes represented by the rectangles, scene i is highlighted to show a sample environment. This has multiple interaction points symbolized as stars, such as conversational agents offering story comments. Interactions can change over time based on the story context. Automatic exploration within each scene is enabled by training the avatar with reinforcement learning.

The structure shown in Fig. 2 also works for interactive films with non-linear narratives. These can be supported by re-sequencing the timeline according to user actions during branch (or choice)

selection; story branches can be loaded in different orders to present alternate events. In addition, this technique can be extended for non-animated or live-action films; 3D scenes can be captured during content creation with multiple camera lens, to later reconstruct a 3D environment. An example option is 360-degree photographs mapped to skyboxes [17].

3.2 3D Environment Exploration

Scenes in the story are modeled as 3D environments. Typically, a static, empty scene is used as a base. When the user enters a scene, they are represented by an avatar placed at a set spawn point. Our examples position the avatar as an entity outside the video’s main characters, so that it may freely explore scenes without disrupting character consistency. However, other stories could be designed to use the video protagonist as the avatar. We refer to entities in the original video as *characters* and their representations in the 3D scene as conversational *agents*. In our system, selected inanimate objects in the background of the video can also “come to life” in the 3D scene as agents. The 3D scene is populated with agents and interactions according to the current video frame. While the base scene is unlikely to change often, interactions within it can be updated separately to reflect story progress. We now outline three types of interactions motivating exploration of the 3D scene, which can be expanded in the future.

3.2.1 Environmental Interactions. The environment can be examined or manipulated by attaching scripted animations to particular points in the scene. This is convenient to include secrets or background information. For example, interacting with a book on a desk may zoom on its contents, revealing a password. Other interactions can simply be for fun, like flickering a light to surprise agents. Lastly, some interactions could affect the story; suppose there is a plot point where a character hides inside a room. If the user avatar does not interact, the character will hide in a closet and be found by default. But, if the avatar pulls the carpet aside, it reveals a trap door that the character spots to avoid detection.

3.2.2 Conversations. Conversational agents representing story characters or objects can be placed around the environment. Allowing the avatar to speak with various character or object agents can illuminate different perspectives and offer world-building, foreshadowing, or amusing remarks. The deployment of agents and conversational topics depends on the current scene. Agents are set up by positioning them in the scene following the characters’ situation in the story. Dialogue is attached with optional points where the user avatar is invited to respond (e.g. with a menu). Dialogue can be written manually by story authors or be automatically generated [19]. Each dialogue interaction would remain available over particular time ranges, changing after key plot events.

Conversations can be one-way or two-way based on context. Consider a story where the user avatar is a ghost; one character is a child who sees the ghost as an imaginary friend. The avatar can speak with the child agent, potentially influencing their actions. The user avatar can also talk with objects who come to life as the child’s imaginary friends. These object agents provide background on the setting and give guidance. For the other human agents who cannot see the avatar, the avatar can instead listen to their private

comments to learn about their perspectives on story events. This is useful to flesh out characters who receive less screen-time without altering the story flow.

3.2.3 Game-like Interactions. While the use of an avatar and 3D environments are already a form of gamification [9], we now consider game-like interactions where the user must complete some objective for a reward. Agents can prompt the user to fulfill a task or mini-game in exchange for new information or unlocking an alternate branch. Different task patterns can be designed and customized with different models to relate the task to the story. Consider a scene with a broken locket that an agent asks the user to fix. The user can find the locket’s parts by searching the scene, which encourages exploration. This pattern can be reused by scattering other items in the environment, like clues to a puzzle. This task only uses the original scene, but mini-games can also bring up a custom interface. For example, the user may be asked to guess where a key item is located under a set of boxes. The boxes are overlaid atop the regular view so that the user can select one of them. Obtaining this item can hint at future story outcomes or let the user unlock a different path. By producing different patterns, the possible interactions offered by the 3D scene are highly flexible.

3.3 Avatar Automatic Exploration

For lean-back users, watching the automatic exploration of the avatar is an effective alternative to manual control. In this mode, walking around and interacting with the scene is handled by the avatar without user interference. We map the problem of automatically guiding the avatar to all interactions in a 3D scene, to a reinforcement learning agent’s action-deciding problem [23]; the avatar is considered as the agent, and its possible actions are 4-directional movements, i.e. left, right, up and down. The environment provides reward to the agent, influencing which direction it moves next.

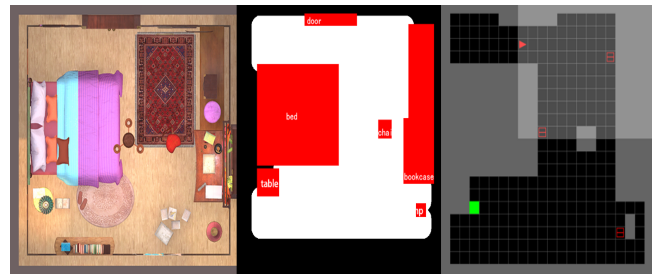


Figure 3: Left: the top-view of an example 3D scene; middle: the floor plan; right: the corresponding MiniGrid environment.

Fig. 3 presents the original top-view image of a 3D scene with its associated floor plan. On the right, a mapped MiniGrid environment (generated by the Minimalistic Gridworld Environment tool [7]) illustrates how the 3D scene is converted to a 2D reinforcement learning environment. In the MiniGrid map, the red triangle represents the avatar, red squares represent keys to acquire, and the green square represents the destination. The avatar can walk over black tiles, but gray tiles are blocked due to physical obstacles. Keys represent interaction points in the gamification. The destination

represents the ending point of the exploration. The keys can be activated automatically when the avatar is in close proximity or a required task has been completed. The avatar is trained over many iterations to maximize reward; this is done by interacting with all keys before reaching the destination, in minimal steps. We use the Advantage Actor-Critic (A2C) algorithm [14] for training the avatar and define reward, r as:

$$r = 1 - \alpha \cdot \frac{s}{M} + \beta \cdot k \quad (1)$$

The first two terms motivate quick exploration by providing decreasing reward as the number of steps taken by the avatar increases. The last term grants extra reward when the avatar finds keys. In the equation, s is the current steps taken, M is the maximum steps allowed (by default, $M = 4L^2$, where L is the length of the environment in number of tiles), α is a coefficient between 0 and 1 impacting the sensitivity of steps taken on the reward, β is the amount of bonus reward (valued between 0 and 1) given per key, and k is the number of keys collected. We experimented on environments containing 3 keys, with areas up to 25x25 tiles. α was set to its default value of 0.9, while β was tested with values between 0 to 1. The β value of 0.5 was found to yield low training times and correct behavior. The training on each environment can be conducted offline. Trained models are stored in a database to be invoked for online path planning. Because the avatar can start routing at random positions, automatic exploration can be enabled anytime.

4 EVALUATION

To test our model, we created a short 3D animation starring a girl named Ella. Character models (supporting 4-directional movements and lip syncing) and environments were created with *Maya* and *3ds Max* [1, 2]. Our prototype system was developed in the *Unity* engine and can handle both 3D rendering and 2D video playback [26]. The system is controlled by the user with a TV remote, but can be generalized to other modes of interaction (e.g. touchscreen, VR) as future work. By pressing a specific button at any time, the user switches between the 2D video and the 3D environment. The video uses scripted character animations and camera angles captured from the 3D scene, allowing seamless switching to 3D while maintaining character positions. Upon switching, the avatar and interactions tied to the current video frame are loaded to the scene. TV remote controls are used for navigation and camera adjustment. A single button is programmed to let users interact with points of interest.

Our prototype implements the features discussed in Sec. 3.2 - an environmental interaction of finding a clue in a book, one-way and two-way conversations with object agents on story events, and two game-like tasks following the patterns in Sec. 3.2.3. Currently, only objects have dialogue, since other agents have more complex positioning to be addressed. Additionally, we included the feature of switching viewpoints to other characters (e.g. Ella's first-person view) during the main video. This was to compare scene interactions with a more lean-back form of engagement.

We conducted interviews by showing videos of the prototype. All users first viewed the introduction of the film normally. The next videos demonstrated switching to different interactions before resuming the story. To gauge the value of each feature, we separated

them from general 3D exploration when possible. For example, speaking with objects was tested by presenting a 2D view of the object, while the avatar could not move. After each video, users were asked a mixture of structured multiple-choice and open-ended questions. Additional discussion was prompted when appropriate. Due to the restrictions of the Covid-19 pandemic, we interviewed all participants remotely with *Zoom* [31]. Interviews were recorded and typically lasted 30-45 minutes. A total of 38 Americans (19 M, 18 F, 1 unspecified) were recruited through an online listing and compensated \$20 USD. We balanced the number of users in the age groups of 8-17, 18-29 and 30-45 years. The mean age was 25.0 years with a standard deviation of 11.1. We now present and discuss our findings regarding the value of anytime switching, exploratory interactions, and avatar automation.

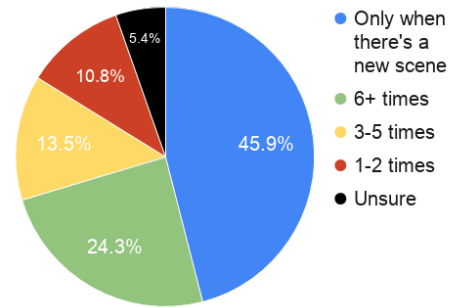


Figure 4: How often users would explore the 3D environment in a 20-minute video.

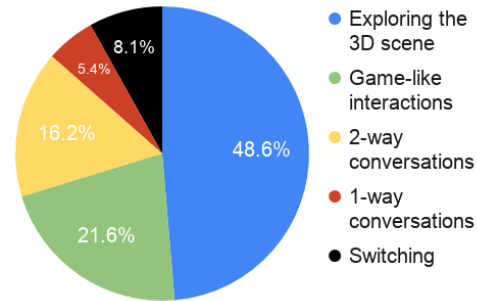


Figure 5: A poll of users' favorite features.

4.1 Value of anytime switching

Users responded positively to the freedom provided by the anytime switching feature, compared to other interactive films (*"In this one you can freely explore and pause whenever you want. In Netflix you have a timer", "Gives me more control"*). To observe how users varied in their degree of interaction, we asked how often they would interact during a 20-minute film. Around 80% of users said they would speak with objects either 1-2 times or 3-5 times in near even proportions. For general exploration, Fig. 4 shows that 46% of

users would only explore when there is a new environment, and the remaining users were split. This illustrates that users have a mixture of interactive needs, which are naturally supported by the anytime switching of this model.

4.2 Value of scene interactions

When we asked users to compare features, we found that environmental interactions were the most appealing. Fig. 5 shows that when asked to choose only one feature, almost half the users chose "exploring the 3D scene"; this included looking around freely and opening a book. Users enjoyed the book since it gave a hint related to the story. For other environmental interactions, Fig. 6 shows that most users wanted to find hidden secrets, and over half also said they would explore for fun and to affect the story. Additionally, over 60% of users felt that merely examining the scene, even without interaction, was valuable. They explained that *"the artists that created this setting put in a lot of work that you don't normally get to see [...] seeing details like this can really enhance the storytelling"*. The remainder felt that the story value depended on how the environment was designed and if they had a purpose for exploration. The 40% of users who were unsure about the environment's value were asked if the exploration feature should be removed if it does not add to the story. 80% of this subset said they would still want the option for novelty and entertainment. They reasoned that *"even if it's not important to the story, it's the most natural way of interacting with the environment"*.

For conversations, users felt that talking to object agents was beneficial. But, they specified that the value depended on the information given; some felt that non-story-related comments were wasteful. They were not concerned about objects spoiling the story, and said *"having some secret info gives you a reason to talk to the objects"*, noting that they *"wouldn't want to miss anything really important. It should just be extra facts"*.

For game-like interactions, users were first shown a guessing game to obtain an item implied to aid Ella later. This was to ascertain the worth of games that give story hints. Fig. 7 shows that 70% of users felt this interaction was helpful because of its foreshadowing and letting the user influence future events. Users commented that they *"think it's fun to do games in the middle of a movie, gives info and something to explore"* and *"It feels more like you're in the story and can affect it, like it's a game and not watching a TV show"*. However, the others felt the game's connection to the current scene was unclear, and recommended moving the game closer to where it would be relevant.

We compared this result to the second game, which encourages the user to explore the environment for a specific item. 76% of users preferred this task over the previous game due to the novelty of the 3D environment. The remainder preferred the first game because its story connection was more clear. Overall, Fig. 5 shows that game-like interactions were the second-most popular feature. We conclude that most users enjoy gamification of the environment, since it gives a clear direction for influencing the story and is rewarding (*"Sometimes when you explore freely, it can seem aimless [...] I like having a goal, like in a game"*, *"I want to have some involvement in Ella's decisions [...] after I do something, I can go back and see how the setting changed"*).

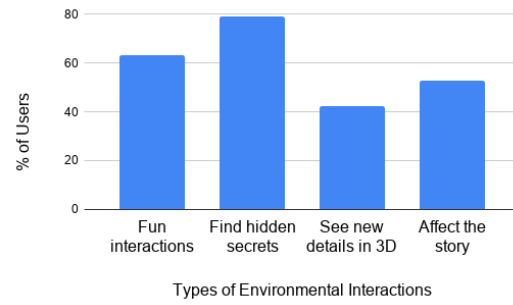


Figure 6: Preferred types of environmental interactions.

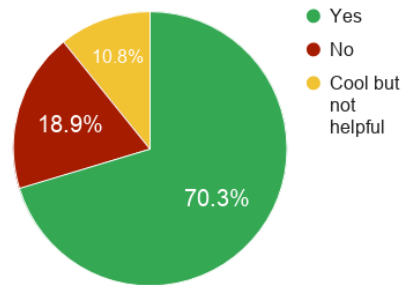


Figure 7: Whether users considered the guessing mini-game to be helpful for their story viewing.

In general, active interactions were preferred over the passive features of switching viewpoints and 1-way comments. Users were enthusiastic about the diverse interactions offered by the environment to drive narrative outcomes (*"In Netflix films, the choices are 'either-or'. Here, you can try a lot of different things, it's very open"*). When asked about other ways they would like to interact with objects, 70% of users mentioned functional purposes like pouring tea with a kettle. Often, this suggestion was coupled with the notion of taking the object to be used later. Users felt that carried objects could give them guidance or hints for solving puzzles, so they were positive about adding more game-like mechanics.

4.3 Value of automation

When asked if they preferred manual avatar control or automatic exploration, only 18% of users chose the latter. However, a third of the remaining users said they would still like the option for automation depending on their mood. Active users liked deciding what to engage with to avoid undesired interactions. Conversely, passive users felt that manual control could require too much effort (*"Controlling the walking might be too troublesome"*, *"I'm usually in the mood to just click around rather than explore"*). Those who wanted both options said automation would be useful to relax (*"I think it depends on the time of day. If it's late at night and I'm tired, I'd rather just watch the movie. [...] But if I feel more energetic, then I'd try to explore"*, *"If I play a lot of interactive movies, then I*

need to rest afterward”). Considering both passive users and those with shifting needs, around half of our sample would benefit from optional automation.

5 CONCLUSIONS

In this paper, we presented an environment-focused model for a new interactive TV experience. At any time, users are invited to join the 3D environment of the story. They can discover hidden details, engage with characters through conversations or exploration, and complete game-like tasks to impact the story development. The user can also choose a lean-back experience, watching their avatar self-explore the 3D environment of specific scenes that interest them. This is the first lean-back-compatible TV model offering a completely immersive experience driven by users, illuminating a new direction for designing interactive videos appealing to a wide audience. User interviews on our prototype confirm the significant value of environmental exploration and gamification for TV audiences. Inspired by our results, our next step will focus on designing the 3D environment to enhance narratives by tailoring interactions to the story progress, motivating exploration with hidden secrets, and emphasizing engagement the first time a scene appears. We will also investigate more gamification features like puzzles to offer users deeper interactions and exploration.

ACKNOWLEDGMENTS

We gratefully thank the creative design and engineering team of TCL Research America, including Allan He, Weimin Ding, Chen Mi, Weiqiang Lei, Jianliang Yi, and many others without whom this project could not be realized. We also thank Mitacs for supporting this work, and all research participants for their time.

REFERENCES

- [1] Autodesk. 2021. 3ds Max. <https://www.autodesk.com/products/3ds-max/overview>
- [2] Autodesk. 2021. Maya. <https://www.autodesk.com/products/maya/overview>
- [3] Sojung Bahng, Ryan M. Kelly, and Jon McCormack. 2020. Reflexive VR Storytelling Design Beyond Immersion: Facilitating Self-Reflection on Death and Loneliness. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI'20)*. Association for Computing Machinery, Honolulu, HI, USA, 1–13. <https://doi.org/10.1145/3313831.3376582>
- [4] Charlie Brooker and David Slade. 2018. Black Mirror: Bandersnatch. <https://www.netflix.com/ca/title/80988062>
- [5] Marc Cavazza, Fred Charles, and Steven Mead. 2002. Interacting with virtual characters in interactive storytelling. In *Proceedings of the 1st International Joint Conference on Autonomous agents and multiagent systems*. Association for Computing Machinery, New York, NY, USA, 318–325. <https://doi.org/10.1145/544741.544819>
- [6] Umer Asghar Chattha, Uzair Iqbal Janjua, Fozia Anwar, Tahir Mustafa Madni, Muhammad Faisal Cheema, and Sana Iqbal Janjua. 2020. Motion Sickness in Virtual Reality: An Empirical Evaluation. *IEEE Access* 8 (July 2020), 130486–130499. <https://doi.org/10.1109/ACCESS.2020.3007076>
- [7] Maxime Chevalier-Boisvert, Lucas Willems, and Suman Pal. 2018. Minimalistic Gridworld Environment for OpenAI Gym. <https://github.com/maximecb/gym-minigrid>
- [8] Jahee Cho, Atit Kothari, Zixu Ding, Yeongmin Won, Stephanie Fawaz, and Xu Cheng. 2016. Injustice: Interactive Live Action Virtual Reality Experience. In *ACM SIGGRAPH 2016 VR Village (Anaheim, California) (SIGGRAPH '16)*. Association for Computing Machinery, New York, NY, USA, Article 9, 2 pages. <https://doi.org/10.1145/2929490.2929493>
- [9] Sebastian Deterding, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011. From Game Design Elements to Gamefulness: Defining Gamification. In *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments, MindTrek 2011 (Tampere, Finland)*, Vol. 11. Association for Computing Machinery, New York, NY, USA, 9–15. <https://doi.org/10.1145/2181037.2181040>
- [10] Saumya Gupta, Theresa Jean Tanenbaum, Meena Devii Muralikumar, and Aparajita S. Marathe. 2020. Investigating Roleplaying and Identity Transformation in a Virtual Reality Narrative Experience. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI'20)*. Association for Computing Machinery, Honolulu, HI, USA, 1–13. <https://doi.org/10.1145/3313831.3376762>
- [11] Jacek Jankowski and Martin Hachet. 2015. Advances in Interaction with 3D Environments. *Computer Graphics Forum* 34, 1 (2015), 152–190. <https://doi.org/10.1111/cgf.12466>
- [12] Kirsten Johnson. 2008. Lost Cause, an Interactive Film Project. In *Proceedings of the 16th ACM International Conference on Multimedia (Vancouver, British Columbia, Canada) (MM '08)*. Association for Computing Machinery, New York, NY, USA, 569–578. <https://doi.org/10.1145/1459359.1459435>
- [13] Jean-Luc Lugrin, Marc Cavazza, David Pizzi, Thirid Vogt, and Elisabeth Andre. 2010. Exploring the usability of immersive interactive storytelling. In *Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST)*. Association for Computing Machinery, Hong Kong, China, 103–110. <https://doi.org/10.1145/1889863.1889887>
- [14] Volodymyr Mnih, Adria Puigdomenech Badia, Mehdi Mirza, Alex Graves, Timothy Lillicrap, Tim Harley, David Silver, and Koray Kavukcuoglu. 2016. Asynchronous methods for deep reinforcement learning. In *International conference on machine learning*. Journal of Machine Learning Research, New York, NY, USA, 1928–1937.
- [15] Nick Montfort and Emily Short. 2012. *Interactive Fiction Communities From Preservation through Promotion and Beyond*. Dichtung Digital. Retrieved Dec. 26, 2020 from www.dichtung-digital.org/2012/41/montfort-short.htm
- [16] Oculus Story Studio. 2015. Henry. <https://www.oculus.com/story-studio/films/henry/>
- [17] Dale Patterson. 2018. 360 Degree Photographic Imagery for VR: Challenges and User Experiences. In *Proceedings of the Australasian Computer Science Week Multiconference (Brisband, Queensland, Australia) (ACSW '18)*. Association for Computing Machinery, New York, NY, USA, Article 44, 7 pages. <https://doi.org/10.1145/3167918.3167955>
- [18] Bernard Perron, Dominic Arsenault, Martin Picard, and Carl Therrien. 2008. Methodological questions in 'interactive film studies'. *New Review of Film and Television Studies* 6, 3 (Dec. 2008), 233–252. <https://doi.org/10.1080/17400300802418552>
- [19] Grant Pickett, Foad Khosmood, and Allan Fowler. 2015. Automated Generation of Conversational Non Player Characters. In *Eleventh Artificial Intelligence and Interactive Digital Entertainment Conference (California, CA, USA) (AIIDE-15)*. AAAI Press, California, CA, USA, 92–98. <https://www.aaai.org/ocs/index.php/AIIDE/AIIDE15/paper/view/11608>
- [20] Oneris Daniel Rico Garcia, Benjamin Tag, Naohisa Ohta, and Kazunori Sugiura. 2017. Seamless Multithread Films in Virtual Reality. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction (Yokohama, Japan) (TEI '17)*. Association for Computing Machinery, New York, NY, USA, 641–646. <https://doi.org/10.1145/3024969.3025096>
- [21] Christian Roth and Hartmut Koenitz. 2019. Bandersnatch, Yea or Nay? Reception and User Experience of an Interactive Digital Narrative Video. In *Proceedings of the 2019 ACM International Conference on Interactive Experiences for TV and Online Video (New York, NY, USA) (TVX'19)*. Association for Computing Machinery, New York, NY, USA, 247–254. <https://doi.org/10.1145/3317697.3325124>
- [22] Supermassive Games. 2015. Until Dawn. <https://www.supermassivegames.com/games/until-dawn>
- [23] Richard S. Sutton and Andrew G. Barto. 2018. *Reinforcement learning: An introduction*. MIT Press, Cambridge, MA, USA.
- [24] TellTale Games. 2013. The Wolf Among Us. <https://www.telltale.com/the-wolf-among-us/>
- [25] The Fullbright Company. 2013. Gone Home. <https://gonehome.game/>
- [26] Unity Technologies. 2021. Unity. <https://unity.com/>
- [27] Haohong Wang. 2019. Device and method for interactive video presentation. US patent pending, Aug. 2020.
- [28] Haohong Wang. 2019. Wishing TV. US patent pending, Dec. 2019.
- [29] Tobias Weber. 2017. Late Shift. <https://lateshift-movie.com/>
- [30] Birk Weiberg. 2002. Beyond Interactive Cinema. Retrieved Sept 13, 2020 from <http://www.keyframe.org/txt/interact/>
- [31] Zoom Video Communications Inc. 2021. Zoom. <https://zoom.us/>